Sales Fraud Detection Analysis

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Abstract

This paper will investigate whether the IsolatedForest unsupervised machine learning method can be applied across sales transactions reported by sales representatives to determine if product sales are misreported. We will evaluate the anomaly detection methods in a supervised manner with the test data using a binary classification of the labeled data.

Keywords: Isolation Forest, Binary Classification, Precision, Recall, Anomaly Detection Supervised Learning, Fraud Detection, Sales Transactions

Introduction

Fraud is committed when sales representatives misreport their product sales with higher numbers to earn a higher commission. The management team needs a fraud detection method in order to determine whether a sales representative committed fraud. We will use the Isolation Forest across unlabeled sales transactions across 800 products. Our training data consists of nearly 134 thousand unlabeled sales transactions which will be classified into either fraudulent (an anomaly) or non-fraudulent (normal). Our results will be compared against manually inspected sales transactions. Given the unbalanced nature of the test data, we will utilize precision and recall (like the F1 score) as evaluation metrics. The anomaly detection method can be applied to various scenarios that mimic the data structure as the sales transaction data.

Literature Review

Isolation forests is a common method for fraud detection across a variety of applications. One example was the application in fraud detection of Credit Card transactions (Waspada, Bahtiar, Wirawan, Awan 2023). Isolation forest, an unsupervised algorithm, is efficient in detecting anomalies. The authors improved the method by implementing additional techniques focused on the amount of training data, feature selection, amount of fraud detection, and setting hyper-parameters. They validated their results by comparing the precision, recall, f1, and ROC-AUC, and AUCPR scores. Given the success of these studies, we are hopeful that we can assess the success of our fraud detection on sales transactions by using either methods. As discussed later in this report, the IsolationForest method is more efficient with a sparse matrix compared to a Principal Components Analysis approach (Scikit-learn, n.d.)

Methods

Firstly, as is standard within our classes, we conduct an Exploratory Data Analysis (EDA) on the data, including a decision on how to handle null entries. As there were no more than 3.3% (in Quant) so we determined the null entries could be removed from the analysis. Then we answer some basic questions, running through averages for the sales reps, average total sales per product, and average sales price per product to see how the various reps compare amongst each other.

Next, we would normally move to a bit of feature engineering. With the variables being mostly categorical and numerical features reasonably distributed, we instead move to model tuning. We did add in the difference between the sales rep's average prices and the overall average price for a given product, called Price_dev. The categorical variables are translated to dummy coded variables as is standard practice. This ends up being about 5800 variables, which is more than we'd like to work with, so we move to truncated singular value decomposition instead of PCA as SVD is more efficient for sparse matrices. Utilizing a scree plot, we could limit the number of components while still getting a majority of the variance explained within those components.

As for the modeling itself, we started with an isolation forest which essentially partitions the data based on entropy or disorder, working to identify the points that split from the pack the fastest. However, with a predetermined amount of data to be labeled anomalous (the parameter called contamination), which originally was set to 5%. However as we have test data we can set contamination to be what's expected based on the raw data provided. In this case, about 8%.

To test the accuracy, we ran the same isolation forest method 5 times with 3 random seeds (7, 19, 928) to ensure reliable results. Finally, we can run our trained models against the testing data to compare what our models found as anomalies versus confirmed anomalies. After a bit of playing around we settled on this original ending as our best version for the models.

Results

Our first check into averages for the Quant and Val variables per sales rep showed multiple outliers on the top end (higher selling vs other reps) as shown in the first set of figures in the Appendix. While these outliers may just be overperformers, there is a chance some of their sales are coming from fraudulent entries for their sales record. Checking into the same averages per product, we see that there are outliers at the top end, which similarly may be the result of anomalistic entries.

Once we had our features created, the initial scree plot search of <=50 entries only explained about 17.5% of the variance. Therefore we went with one up to 1000 as shown in Figure 5, which ended up explaining about 75% of the variance. This is deemed sufficient within this study, with about 17% of the variables needed moving forward.

After training the isolation forest model with the three seeds, the F1 is determined to be about 91.802% correctly identified as anomalies aka fraudulent entries. The table in the Appendix shows what altering the number of estimators and components does to the F1 results, with a cap of about 92.546% where there are 100 estimators and 1000 components. Overall, we were successfully able to label anomalies with greater than 90% accuracy.

Conclusions

This study proves that it is possible to quickly put together a model that can assess a sale and determine whether it may be fraudulent. Each sale may need to be investigated to ensure it’s not just an unusually big order or has some factor changing

the price compared to a normal sale. If many of these sales reside with one sales rep, it is likely they are intentionally completing these sales in a fraudulent manner and should be investigated and potentially let go by the business. While this may be the result of an improperly trained employee who is entering in their sales incorrectly, this would then also allow the client to look for a similarity among these employees like a specific work unit or shared trainer that is leading to these incorrect entries.

References

Scikit-learn. (n.d.). *IsolationForest*. Scikit-Learn.org; Scikit-learn. Retrieved June 8, 2025, from https://scikit-learn.org/stable/modules/generated/sklearn.ensemble.IsolationForest.html

Waspada, I., Bahtiar, N., Wirawan, P. W., & Awan, B. D. A. (2020). Performance Analysis of Isolation Forest Algorithm in Fraud Detection of Credit Card Transactions. Khazanah Informatika : Jurnal Ilmu Komputer Dan Informatika, 6(2). https://doi.org/10.23917/khif.v6i2.10520

Appendix

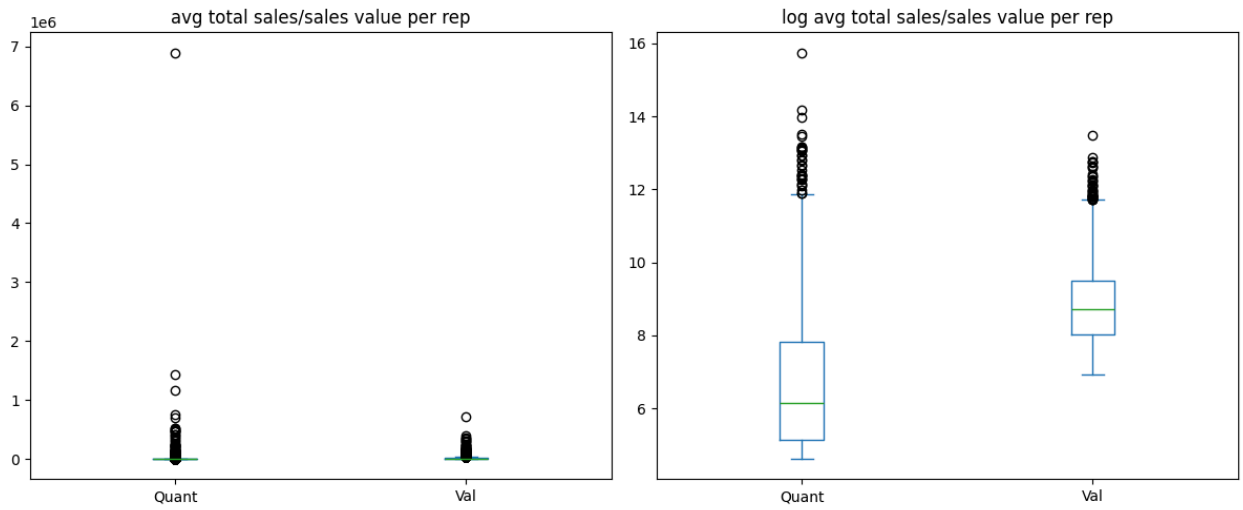

Figure 1 and 2. Average Quant and Val for each sales rep and the log transformation of the same information.

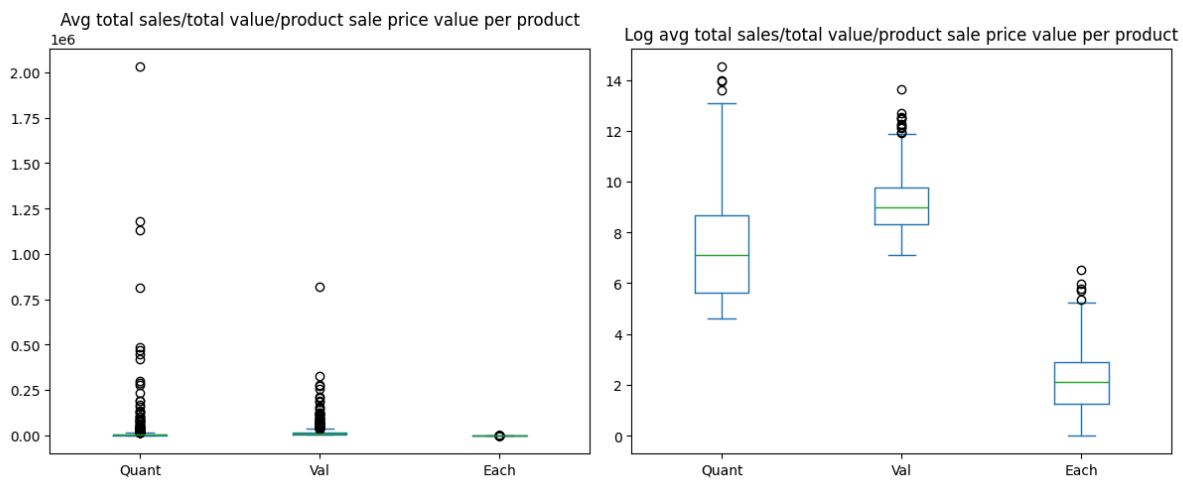

Figure 3 and 4. Average Quant, Val, and Each (Val/Quant) for each product as well as a log transform for visibility.

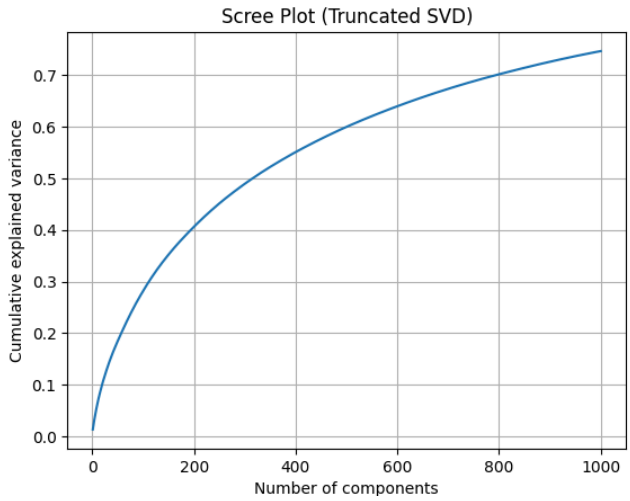

Figure 5. Scree plot for the features' variance explanation within the truncated SVD training.

	estimators	components	f1_score
0	100	200	0.882861
1	300	200	0.889392
2	1000	200	0.894366
3	100	400	0.913292
4	300	400	0.888801
5	1000	400	0.897704
6	100	800	0.912916
7	300	800	0.902941
8	1000	800	0.905843
0	100	1000	0.925459
1	300	1000	0.915164
2	1000	1000	0.913240
3	100	1500	0.924205
4	300	1500	0.916620
5	1000	1500	0.917280
6	100	2000	0.923012
7	300	2000	0.918447
8	1000	2000	0.922586

Table 1. A table of F1 scores for the various estimators and components for the IsolatedForest model.